

BARE METAL / SHARP X68000

PONG

C WITHOUT HUMAN68K
1P / 2P - KEYBOARD /
GAMEPAD

elf2x68k + IOCS + mode 12

Makefile -> dist/pong.xdf



GameContext unifies application and mode state

ApplicationState

- old_mode
Display mode before launch
- mode : GameModeld
Current screen state
- application_loop()
Owns all GameMode transitions

GameContext

application : ApplicationState

title : TitleState

pong : PongGameState

winner : WinnerState

One shared context holds all state

make builds the bare-metal XDF end to end

01 /
COMPILE

m68k-xelf-gcc

-m68000 -O2
-specs=x68knodos.specs
Link address 0x6800

02 /
BOOT

Generate human.sys

Link directly as human.sys.
No Human68k system files
required.

03 / XDF

dist/pong.xdf

Create a new XDF containing
only human.sys.

512 x 480 display at 60 Hz

MODE

SYNC

TIMING

CRTMOD(12)

Standard 31 kHz mode:
512 x 512 / 65,536 colors.

wait_vdisp()

Wait for V-DISP on MFP GPIF;
one update per frame.

R05/R06/R07 -> R04

Set display position first;
set 525 scan lines last.

_ONTIME prevents lockups with a wait timeout

GameMode standardizes each screen lifecycle

```
typedef struct {
    void (*initialize)(GameContext *);
    GameModeId (*update)(GameContext *);
    void (*finalize)(GameContext *);
} GameMode;

next =
game_modes[current].update(context);
if (next != current) {
    game_modes[current].finalize(context);
    game_modes[next].initialize(context);
}
```

application_loop() owns every transition

- 01 initialize() enters the screen
- 02 update() processes one frame
- 03 Return the next GameModeId
- 04 finalize() leaves the screen
- 05 title / pong / winner share one interface

Add a GameMode with state, three functions, and one table row.

Keyboard and two gamepads supported

PLAYER 1
W / S
PAD 1 UP / DOWN

PLAYER 2
CURSOR UP / DOWN
PAD 2 UP / DOWN

TITLE

UP / DOWN SELECT - RETURN / SPACE / A
START

MATCH

Q / ESC returns to title

STATUS KEYBOARD + 2 PADS

Public repository organized by purpose

01

src/

pong.c and Makefile.
Source and build definitions.

02

dist/

Generated boot disk:
pong.xdf.

03

docs/ + README

Technical report: pong-en.pdf.
README covers overview and
build.